A Multi-Agent Academic Helpdesk for IIT Jodhpur

CSL7650 Autonomous Systems — Assignment 2

[TODO: team names and roll numbers]

October 2026

Abstract

We build a local, laptop-runnable academic helpdesk that answers questions about IIT Jodhpur programmes, rules and regulations, course slots and the timetable, drafts (but never sends) e-mails to instructors, cites the document every answer comes from, and refuses and logs questions it cannot answer. The system uses Google ADK with 8B models served by Ollama (Qwen3-8B, Llama-3.1-8B) through LiteLLM, retrieval-augmented generation over 24 institute documents (1267 chunks, `nomic-embed-text` + ChromaDB) and deterministic tools over tables extracted from the official slot-system and classroom-allocation PDFs. A single-agent baseline is compared with a multi-agent organisation (an orchestrator delegating to Policy, Schedule and Email agents) on 30 questions with repeated runs. The multi-agent system with Qwen3-8B reaches TBD accuracy (single agent: TBD) and a refusal F1 of TBD. **[TODO: one sentence on the main finding once the full evaluation has been run]**

1 Problem and environment

The helpdesk must answer questions such as "What is the attendance requirement to sit in the Major exam?", "What is the slot for Autonomous Systems?", "How many 400-level electives has the Humanities department floated this semester?", "How many Open Elective credits do I need for a B.Tech.?" and "Write an e-mail to the instructor of CSL7570 that I need leave for 2 days", and must say it does not know for questions such as "What are the dining options on campus?".

Using the environment properties from the course, the helpdesk's environment is *inaccessible* (the model sees only the retrieved passages, never the whole corpus), *non-deterministic* (sampling makes answers vary between runs — "from an agent's perspective even a deterministic environment is non-deterministic", which is why every question is run several times), *episodic* (questions are independent), *dynamic* (rules, slots and office orders change outside the agent; re-ingestion is how the system adapts) and *discrete* (a finite set of routes and tools).

2 Design principles from the course

Table 1 maps the concepts taught in class to concrete design decisions; the subsections expand the most important ones.

2.1 The helpdesk as an organisation

We follow the organisation-centric view: the helpdesk is an entity whose heterogeneous agents together fulfil one goal (correct, cited help), so no agent pursues a private goal. Roles are written in the *role description* format from class (objective, rights, norms), Table 2. The dependency type between orchestrator and specialists is *delegate*.

2.2 Institution and norms

The guardrails implement the institution's "rules of the game". With the normative system $N = \langle E, F, G, C, \Delta \rangle$ from class: exogenous events are student messages; the generation relation G maps a

Table 1: Course concepts and where they appear in the system.

Concept	Realisation in the helpdesk
No full autonomy: constraints and guardrails	Scope governor, citation check, draft-only e-mail, escalation log
Effectoric capability; action pre-conditions	Each agent's tool set is its effectoric capability; an e-mail is drafted only after the instructor is resolved; an answer only after retrieval succeeds
Reactive vs. proactive balance; subsumption	A reactive keyword governor (highest-priority layer) subsumes deliberation for clearly off-topic questions; goal-directed agents handle the rest
BDI: `brf`, `options`, `filter`, `execute`	Retrieval updates beliefs; candidate routes are desires; the orchestrator's delegation is the intention; tool calls execute it; the pending e-mail persists across turns
Horizontal layering (mediator, m^n bottleneck) vs. vertical two-pass	Delegation to one specialist instead of asking every agent and merging answers
Organisation-centric MAS: social, interaction, normative, communication structure	Roles (Table 2); routing graph; norms (cite, do not guess, refuse and log); structured tool results and shared session state
Institutions: obligations, permissions, power; normative system $\langle E, F, G, C, \Delta \rangle$	Per-role tool permissions; obligation to cite and to log; only humans close escalations (Sec. 2.2)
Scenes, scene scripts, landmarks; trace	Query and e-mail scenes (Fig. 1); every run is saved as an event trace
Coordination: blackboard, limited communication, scepticism, locally complete results	ADK session state is the blackboard; specialists only talk to the orchestrator; the citation guardrail does not trust uncited answers; specialists return final cited answers
Organisation adaptation ("emergent patterns become norms")	The unanswered-question log is reviewed by humans, who add documents or rules
POMDP belief state; tiger problem ("listen" before acting)	Retrieval scores are evidence; below the threshold the agent "listens" (asks, or refuses) rather than "opens a door"
ReAct / tool use	ADK's reason–act–observe loop with typed function tools
RAG and grounding	Chunked corpus, embeddings, threshold τ, mandatory citations
Agent communication (speech acts, Contract-Net)	Delegation = *request*, tool result = *inform*, refusal = *refuse*; Contract-Net bidding discussed as an alternative router
Argumentation	Resolving conflicting regulations by recency (Sec. 2.3)

message in state X to institutional events — *answer*, *refuse*, *escalate*, or a violation $e \in E_{violation}$ (an off-topic request, or an answer that cites a document never retrieved). The consequence relation C adds fluents: a refusal adds the obligation fluent *obliged(human, answer(q))* (a line in `unanswered.jsonl`); a successful tool call adds the domain fluent *retrieved(doc)*, which the citation check later reads. Permissions are the per-role tool sets (only the Email agent may create drafts); power to close an escalation belongs to humans only. The ADK session state ("s'") runs alongside the conversation ("s").

2.3 Conflicting documents as argumentation

The corpus contains a real conflict. The UG regulations (2019) say a student below 75% attendance may still sit the exams but needs at least a C grade; the Senate circular of 8 July 2024 says examination performance is *no longer linked* to attendance and "supersedes the earlier policy in totality". As an abstract argumentation framework $\langle A, R \rangle$ with $A = \{a_{2019}, a_{2024}\}$ and $R = \{(a_{2024}, a_{2019})\}$, the grounded extension is $\{a_{2024}\}$. The attack relation is stored explicitly (a `supersedes` column in the manifest). When an amending circular and a regulation it supersedes are retrieved together, `search_documents` ranks the circular first and marks the older passages as superseded. Without this, the first test run answered the attendance question from the 2019 rule, even though the newer circular was among the retrieved passages and the prompt asked for the most recent document.

Table 2: Role descriptions (social structure).

Role	Objective	Rights (tools)	Norms
Orchestrator	Route each question to the right specialist	`transfer_to_agent`, `log_unanswered`	Never answer academic questions itself; refuse and log non-academic ones
Policy agent	Answer questions on rules, regulations and curricula	`search_documents`, `log_unanswered`	Search first; cite every fact; prefer the most recent circular
Schedule agent	Answer slot, room, instructor and course-count questions	`get_course_info`, `get_slot_timing`, `list_courses`, `search_documents`	Never count by hand; cite the allocation/slot documents
Email agent	Draft e-mails to instructors	`get_instructor_email`, `draft_email`	Never send; never invent reasons, dates or an instructor
Human (academic office)	Answer escalated questions	Reads `logs/unanswered.jsonl`	Closes escalations; adds documents

3 System

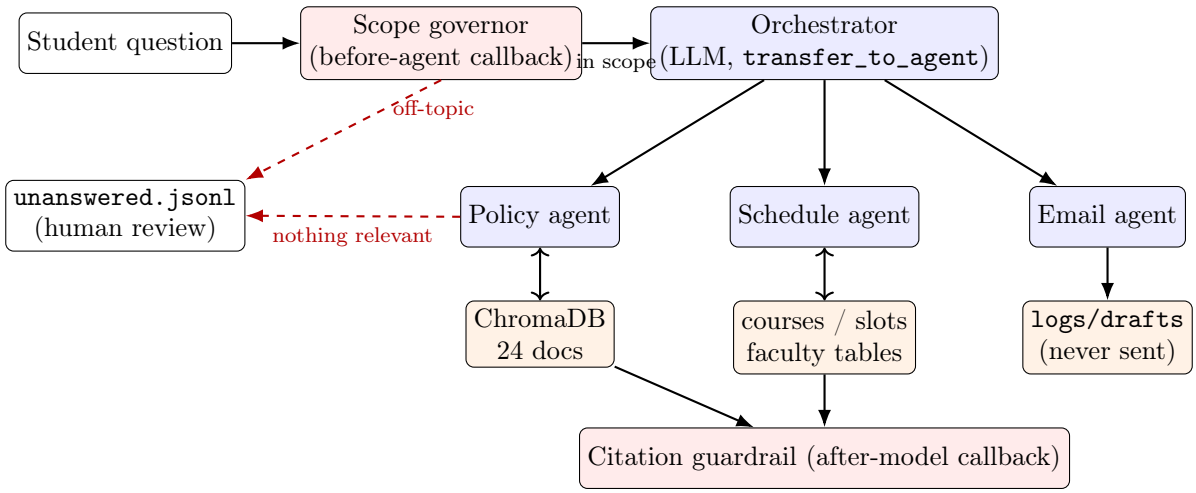


Figure 1: Architecture. Dashed red arrows are refusals that are logged for a human.

Corpus. 24 documents (listed in `ingest/manifest.csv`): UG, PG, M.Tech., M.S. (Research), Ph.D. and joint-programme regulations; the 2024 attendance and examination circulars (scanned, OCR'd with Tesseract); continuation norms; the academic calendar AY 2026-27; the institute statutes; curriculum pages for B.Tech. CSE/AIDE/EE; registration, minor/specialisation and programme-list pages; the slot system and classroom allocation for Semester I 2026-27; a transcription of the slot system's rules; and a faculty directory derived from the allocation with *dummy* e-mail addresses (personal data replaced, as allowed). PDFs are chunked per page, HTML/Markdown per section (800 characters, 150 overlap) and embedded with `nomic-embed-text`.

Tables. The classroom allocation PDF is parsed into 626 course offerings (code, title, L-T-P, credits, slot, department, instructors, room, registrations, page) by reading words in content-stream order (overflowing cells otherwise interleave characters) and splitting instructor cells at backward jumps in x. The slot grid gives 70 (slot, day, time) rows and the faculty table 339 instructors. Because the PDF has no elective column, electives are *inferred* from the slot system's programme-wise allocation

table (HSS courses at level ≥300 and courses in the elective slots of their level); this is a documented heuristic.

Tools. `search_documents` (cosine distance threshold $\tau = 0.35$, top-5, returns `cite_as` strings and document dates), `get_course_info`, `list_courses` (exact counts so the LLM never counts), `get_slot_timing`, `get_instructor_email`, `draft_email` (saves a Markdown draft; never sends; only uses what the student said) and `log_unanswered`. Every tool records the documents it read in the session state.

Guardrails. (1) The scope governor refuses clearly off-topic questions (dining, cab, weather, cricket, ...) before any LLM call; it is deliberately narrow, and anything it misses (e.g. "best way to reach the campus from the city") must be refused by the agents themselves when retrieval finds nothing. (2) The citation guardrail replaces an answer that cites a document no tool retrieved, or that was produced without any tool, by a refusal and logs it; if the model used tools but forgot to cite, the retrieved sources are attached and the run is flagged ("auto-attached").

Threshold τ. On probe questions the best in-scope passage had cosine distance ≤ 0.26 and out-of-scope questions ≥ 0.30, but short keyword queries written by the agents reach about 0.32; we use $\tau = 0.35$ and study 0.28 and 0.45 in the ablation.

4 Evaluation setup

30 questions (Appendix A): 8 policy, 8 schedule, 5 e-mail, 6 out of scope and 3 that sound academic but are not covered by the corpus; all five example questions from the assignment are included. Gold answers were written from the documents and checked against them. Configurations: {multi-agent with LLM orchestrator, single agent} × {Qwen3-8B, Llama-3.1-8B}, 3 runs each at the models' default temperature, on an Intel i5-12450H laptop (CPU only, 16 GB RAM).

Metrics: *accuracy* (required facts present and no refusal; for questions that should be refused, a refusal; automatic checks were verified by hand in `eval/human_labels.csv`); *refusal* precision/recall/F1 and false-refusal rate; over answered in-scope questions, *cited* (the model cited by itself), *correct source* (a cited or attached document is a gold source) and *fabricated* (cites a document never retrieved); *routing accuracy* (first specialist the orchestrator delegated to); latency; and *stability* — questions whose outcome differs between runs.

5 Results

Table 3: Main results (all runs). Citation columns are over answered in-scope questions; routing is the first specialist the orchestrator delegated to.

System	Model	Router	Runs	Acc.	Ref. F1	False ref.	Cited	Correct src.	Fabric.	Routing	Lat. (s)
Not run yet.											

Table 4: Accuracy by question category.

System	Model	Policy	Schedule	Email	Out of scope	Uncovered
Not run yet.						

figure generated after the evaluation run

Figure 2: Accuracy by question category.

Table 5: Questions whose outcome changed across repeated runs (correct in only some runs, or refused in only some runs).

Q	Category	System	Model	Correct	Refusal flip	Text sim.
Not run yet.						

Table 6: Ablation (multi-agent, one run per question): guardrails switched off and retrieval threshold τ changed (default 0.35).

Configuration	Acc.	Ref. F1	False ref.	Correct src.	Fabric.
Not run yet.					

Table 3 gives the main comparison, Table 4 the per-category accuracy, Table 5 the questions whose answers changed across runs and Table 6 the ablation. **[TODO: Describe the findings from the generated tables: single vs. multi agent, Qwen vs. Llama, which questions changed across runs and why, effect of each guardrail and of τ. Write only what the numbers show.]**

5.1 Example transcripts

The following are verbatim outputs of the first multi-agent run. *Not run yet.*

6 Limitations

- Retrieval-distance thresholds overlap between in-scope and out-of-scope questions; the final refusal decision relies on the LLM for borderline cases.
- Elective status is inferred from slot rules; two of 626 allocation rows still contain merged text from overflowing PDF cells.
- E-mail addresses are dummies; lab-slot durations are inferred from the grid layout.
- CPU-only inference makes each question take on the order of a minute or more, which limits the number of runs.
- **[TODO: limitations observed in the evaluation]**

7 Group contributions

[TODO: who did what]

References

[1] G. Weiss (ed.). *Multiagent Systems*, 2nd ed. MIT Press, 2013.

[2] M. Wooldridge. *An Introduction to MultiAgent Systems*, 2nd ed. Wiley, 2009.

[3] R. Brooks. A robust layered control system for a mobile robot. *IEEE J. Robotics and Automation*, 2(1), 1986.

[4] M. E. Bratman. *Intentions, Plans, and Practical Reason*. Harvard University Press, 1987.

[5] P. M. Dung. On the acceptability of arguments and its fundamental role in nonmonotonic reasoning, logic programming and n-person games. *Artificial Intelligence*, 77(2), 1995.

[6] S. Yao et al. ReAct: Synergizing reasoning and acting in language models. ICLR 2023.

[7] P. Lewis et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. NeurIPS 2020.

[8] FIPA. *FIPA ACL Message Structure Specification* (SC00061G), 2002.

A Question set

ID	Category	Question	Expected answer / behaviour	Gold sources
Q01	policy	What is the attendance requirement to sit in the Major exam?	Since the Senate amendment of 8 July 2024, examination performance is no longer linked to attendance, so there is no attendance cut-off for sitting the Major exam. Students are expected to attend fully; attendance should not fall below 75%, and below 75% without a valid reason the instructor can recommend de-registration from the course.	ATTEND_ POL_2024
Q02	policy	How many Open Elective credits do I need to earn in order to get a B.Tech. Degree?	A regular B.Tech. requires 10 Open Elective (OE) credits (Table 11.2: OE = 10 of 150 graded credits; 0 for B.Tech. with specialization).	UG_ CREDIT_ TABLE, UG_ REG_2019
Q03	policy	What is the minimum weightage and duration of the Major examination?	The Major examination is a 3-hour written exam with a weightage of not less than 40% of the overall evaluation.	EXAM_ POL_2024
Q04	policy	I missed my major exam because I was admitted to hospital. How do I apply for a make-up examination?	Apply to the Dean (Academics) through the course instructor within ten days of the missed exam, with the required medical certificate (hostel residents: certificate from the Institute Medical Officer).	UG_REG_ 2019, ACAD_ REG_ PORTAL
Q05	policy	What is the maximum number of credits a UG student can register for in a semester?	A UG student may register for at most 25 credits in a semester (minimum 12).	UG_REG_ 2019, ACAD_ REG_ PORTAL
Q06	policy	How many credits can I take during the summer term?	Students can register for summer courses up to a maximum of eight credits.	UG_REG_ 2019, ACAD_ REG_ PORTAL
Q07	policy	What SGPA does an M.Tech. student need for continuation of assistantship?	SGPA 6.0 (for the first semester, >= 5); continuation in the programme needs CGPA 5.0 (2020 amendment).	CONTINUATION_ NORMS_2020
Q08	policy	How many credits are needed for a minor or interdisciplinary specialization in B.Tech.?	20 credits in the specialization/minor area: 10 can be counted as open electives and 10 are taken as course overload.	UG_REG_ 2019, SPEC_ MINOR_ RULES, ACAD_REG_ PORTAL
Q09	schedule	What is the slot for the course Autonomous Systems?	Autonomous Systems (CSL7650) is in slot E (Tue, Thu, Fri 11:00-11:50), room LHC 106.	CLASS_ ALLOC_2026
Q10	schedule	How many 400 level electives have been floated by the Humanities department in this semester?	Two: LAL4070 Health Economics and LAL4850 Introduction to Climate Change Economics (School of Liberal Arts).	CLASS_ ALLOC_2026

ID	Category	Question	Expected answer / behaviour	Gold sources
Q11	schedule	What are the lecture days and timings for Slot E?	Slot E meets Tuesday, Thursday and Friday, 11:00-11:50.	SLOT_SYS_2026
Q12	schedule	Who is the instructor for Database Systems and in which classroom does it meet?	Database Systems (CSL3050) is taught by Awathare Nitin Niranjan in LHC 2 101 (slot A).	CLASS_ALLOC_2026
Q13	schedule	Who teaches Digital Fabrication and what is its course code?	Digital Fabrication is MEL3050, taught by Gopal Dnyanba Gote (slot A, LHC 205).	CLASS_ALLOC_2026
Q14	schedule	What is the maximum number of students per lab and per tutorial in the slot system?	Maximum 50 students per lab and about 40 per tutorial.	SLOT_SYS_2026, SLOT_SYS_RULES_2026
Q15	schedule	Which slots are used for compulsory courses of UG 2nd year students?	A-E for programme compulsory courses, U for institute electives, and K & N for programme compulsory courses if the strength is below 160.	SLOT_SYS_2026, SLOT_SYS_RULES_2026
Q16	schedule	In which slot and room is Database Systems, and on which days does that slot meet?	CSL3050 Database Systems is in slot A, LHC 2 101; slot A meets Monday, Tuesday and Thursday 09:00-09:50.	CLASS_ALLOC_2026, SLOT_SYS_2026
Q17	email	Write an email to the course instructor of CSL7570 that I need leave for 2 days	CSL7570 is not in the AY 2026-27 Sem I classroom allocation, so the agent must not invent an instructor; it should say the code was not found and ask the student to confirm it (Autonomous Systems is CSL7650).	CLASS_ALLOC_2026
Q18	email	Write an email to the course instructor of CSL7650 that I need leave for 2 days due to fever on 14 and 15 October.	Draft (not sent) to Gaurav Harit, gaurav.harit@iitj-dummy.ac.in, requesting leave for 14-15 October due to fever.	FACULTY_DIRECTORY
Q19	email	Draft an email to the instructor of CSL3050 asking for a make-up exam because I missed the minor exam on 20 September due to hospital admission.	Draft (not sent) to awathare.nitin.niranjan@iitj-dummy.ac.in requesting a make-up for the minor exam missed on 20 September.	FACULTY_DIRECTORY
Q20	email	Draft an email to the instructor of MEL2060 asking about the syllabus for the upcoming quiz.	Draft (not sent) to rajit.ranjan@iitj-dummy.ac.in asking about the quiz syllabus.	FACULTY_DIRECTORY
Q21	email	I need an email drafted to my instructor for a medical absence.	No course is given, so the agent should ask which course (course code) the e-mail is for instead of drafting one.	–
Q22	out of scope	What are the available dining options on the campus?	Refuse ('I don't know') and log the question for a human.	–
Q23	out of scope	What is the best way to reach the campus from Jodhpur city center?	Refuse ('I don't know') and log the question for a human.	–
Q24	out of scope	What is the weather forecast for Jodhpur this weekend?	Refuse ('I don't know') and log the question for a human.	–
Q25	out of scope	Who won the cricket world cup?	Refuse ('I don't know') and log the question for a human.	–
Q26	out of scope	Which movies are showing at the mall in Jodhpur?	Refuse ('I don't know') and log the question for a human.	–

ID	Category	Question	Expected answer / behaviour	Gold sources
Q27	out of scope	Can you write a python script to scrape data from twitter?	Refuse ('I don't know') and log the question for a human.	–
Q28	uncovered	What is the policy for swimming pool access and coach timings for B.Tech students?	Not covered by the documents: say 'I don't know' and log the question.	–
Q29	uncovered	What is the mess fee refund policy if I am on leave for 15 days?	Not covered by the documents: say 'I don't know' and log the question.	–
Q30	uncovered	What are the rules and stipend for the 2029 quantum technology exchange fellowship?	Not covered by the documents: say 'I don't know' and log the question.	–

B Reproduction

```
uv venv --python 3.12 && uv pip install -r requirements.txt
ollama pull qwen3:8b && ollama pull llama3.1:8b && ollama pull nomic-embed-text
python main.py ingest --rebuild
python -m pytest -q
python eval/run_eval.py                       # full grid
python eval/run_eval.py --ablation --model qwen3:8b
python eval/run_eval.py --score-only          # after editing eval/human_labels.csv
cd report && pdflatex report.tex && pdflatex report.tex
```